



Pediatric Tachycardia

Narrow Complex (≤ 0.09 sec)

History

- Past medical history
- Medications or Toxic Ingestion (Aminophylline, Diet pills, Thyroid supplements, Decongestants, Digoxin)
- Drugs (nicotine, cocaine)
- Congenital Heart Disease
- Respiratory Distress
- Syncope or Near Syncope

Signs and Symptoms

- Heart Rate: Child > 180 /bpm
Infant > 220 /bpm
- Pale or Cyanosis
- Diaphoresis
- Tachypnea
- Vomiting
- Hypotension
- Altered Level of Consciousness
- Pulmonary Congestion
- Syncope

Differential

- Heart disease (Congenital)
- Hypo / Hyperthermia
- Hypovolemia or Anemia
- Electrolyte imbalance
- Anxiety / Pain / Emotional stress
- Fever / Infection / Sepsis
- Hypoxia, Hypoglycemia
- Medication / Toxin / Drugs (see HX)
- Pulmonary embolus
- Trauma, Tension Pneumothorax

Assess tachycardia in context of clinical condition
Identify and treat underlying cause of tachycardia

Unstable / Serious Signs and Symptoms
AMS, shock, hypotension
HR Typically > 180 Child
HR Typically > 220 Infant

YES

P	Cardiac Monitor
	Cardioversion Procedure
	0.5 – 1 J / kg Repeat and increase to 2 J / Kg Consider sedation <u>Do NOT delay cardioversion</u> Midazolam 0.1 – 0.2 mg / kg IV / IO / IN May repeat if needed Maximum Single Dose 2 mg Maximum Total Dose 5 mg

NO

B	12 Lead ECG Procedure
P	Cardiac Monitor
	IV or IO Access Protocol UP 6

Single lead ECG able to
diagnose and treat arrhythmia

12 Lead ECG not necessary to diagnose and
treat, but preferred when patient is stable.

Regular Rhythm?

NO

Contact Medical Control

YES

Probable Sinus Tachycardia

Identify and Treat Underlying Cause

Exit to
Age Appropriate
Protocol(s)



Probable SVT

Vagal Maneuvers

Adenosine 0.1 mg / kg IV / IO
Maximum 6 mg

May repeat
Adenosine 0.2 mg / kg IV / IO
Maximum 12 mg

**Notify Destination or
Contact Medical Control**

AT ANY TIME

Pulseless



**Go to
Pediatric Pulseless
Arrest Protocol**



Pediatric Tachycardia

Narrow Complex (≤ 0.09 sec)

Pearls

- **Recommended Exam: Mental Status, Skin, Neck, Lung, Heart, Abdomen, Back, Extremities, Neuro**
- **Monomorphic QRS:**
All QRS complexes in a single lead are similar in shape.
- **Polymorphic QRS:**
QRS complexes in a single lead will change from complex to complex.
- Use length-based or weight-based pediatric resuscitation system for medication, equipment, cardioversion, and defibrillation guidance. Pediatric paddles should be used in children < 10 kg.
- Rhythm should be interpreted in the context of symptoms and pharmacological or electrical treatment given only when symptomatic, otherwise monitor and reassess.
- **12-Lead ECG:**
12-Lead ECG not necessary to diagnose and treat.
Obtain when patient is stable and/or following rhythm conversion.
When administering adenosine, obtaining a continuous 12-Lead can be helpful to physicians.
- **Unstable condition:**
Condition which acutely impairs vital organ function and cardiac arrest may be imminent.
If at any point patient becomes unstable move to unstable arm in algorithm
If IV or IO access is in place, may administer adenosine and repeat, prior to synchronized cardioversion.
- Document all rhythm changes with monitor strips and obtain monitor strips with each therapeutic intervention.
- **Serious Signs and Symptoms:**
Respiratory distress / failure.
Signs of shock / poor perfusion with or without hypotension.
AMS
Sudden collapse with rapid, weak pulse
- **Narrow Complex Tachycardia (≤ 0.09 seconds):**
Sinus tachycardia: P waves present. Variable R-R waves. Infants usually < 220 beats / minute. Children usually < 180 beats / minute.
SVT: $> 90\%$ of children with SVT will have a narrow QRS (≤ 0.09 seconds.) P waves absent or abnormal. R-R waves not variable. Usually abrupt onset. Infants usually > 220 beats / minute. Children usually > 180 beats / minute.
Atrial Flutter / Fibrillation
- **Vagal Maneuvers:**
Breath holding. Blowing a glove into a balloon. Have child blow out "birthday candles" or through an obstructed straw. Infants: May put a bag of ice water over the upper half of the face careful not to occlude the airway.
- Separating the child from the caregiver may worsen the child's clinical condition.
- Monitor for respiratory depression and hypotension associated if Diazepam, Lorazepam, or Midazolam is used.
- Continuous pulse oximetry is required for all SVT Patients if available.